A Study on the Usability of Differential Privacy Libraries

Objective

This study aims to assess your understanding and implementation of differential privacy concepts while using 3 of the most forked and most rated libraries(PyDP, OpenDP, DiffPrivLib) to analyse data. Your participation will provide valuable insights into these libraries' usability and practical application.

Tasks

You will be asked to perform the following tasks using the differential privacy libraries provided:

1. Count the Number of Individuals in Different Income Brackets

- Use differential privacy techniques to count the number of individuals in an income bracket(> 50K) and print the output in below format:

Income Bracket	**Count**
>50K	**NN**

2. Determine the Distribution of Education Levels

- Calculate the average age for each education level within the dataset using differential privacy libraries and print the output.

Education Level	**Average Age**
Graduation	**NN**

3. Fill a Survey

Participation Details

- **Duration**: The study will take approximately 1.5-2 hours to complete.
- **Confidentiality**: All data collected during this experiment will be anonymized and used solely for research purposes.
- **Support**: If you encounter any difficulties or have questions, please contact the study facilitator.

Instructions

- Preparation

- Use the handout provided to familiarize yourself with the concepts of differential privacy if you haven't already referred to the official documentation of the above mentioned Differential Privacy libraries. Ensure you have access to a suitable development environment to run code (e.g., VSCode, Jupyter Notebook, Python IDE).
    - Access the Remote Host via SSH / VNC to perform these tasks.
- Dataset
    - Use the Dataset(downloaded from https://archive.ics.uci.edu/dataset/2/adult) placed under Documents folder on the remote host, which will be used for your analysis.
- Task Execution
    - Implement differential privacy techniques for the specified tasks using 3 libraries. We encourage you to think-aloud during this experiment, so we capture your though process, how you set the privacy budget and other parameters. We will be documenting your reasoning for choosing your coding approach, any parameters, etc.
- Analysis
    - We will analyse the results of your differential privacy implementation to reflect on the effectiveness of your chosen methods and any challenges you encountered.

Assessment Criteria

Your participation in this study will be assessed based on:

- Understanding of Differential Privacy Concepts
    - Demonstrated knowledge of differential privacy principles and their application.
- Implementation Skills
    - Ability to effectively use differential privacy libraries to achieve the desired outcomes.
- Reasoning and Justification
    - Clarity and rationale behind choosing specific privacy budgets and parameters.